

Final Report for Project 1-4

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I. PROJECT 1: ACOUSTIC LINK

A. Stories

Tasks 0 and 1 of Project 1 are both about how we will use the hardware and software tools we are about to use. Task 2 is to generate a signal with two special peaks. The first few tasks are relatively simple, and we can easily complete them. Task 3 is to use sound signals to propagate data between two computers, with a focus on implementing modulation methods, frame structures, and error correction mechanisms. We need to understand the audio interface and generate basic sound waves, then transmit the bit stream, obtain the header of each frame through preamble, and decode the subsequent data. Each of these tasks will face technical and hardware challenges. Although it was very difficult to do this task at the beginning, it greatly deepened our understanding of sound signal transmission after completion.

B. Designs

1) Task 2: Generating Sound Waves at Will

- We are using Python language, and the sampling rate is set to 48000, which is recommended by the task. One reason is that it maximizes the utilization of the audio spectrum. Another reason is that when using other sampling frequencies, the sound card may implicitly perform resampling, thereby introducing additional noise.
- Then we wrote the corresponding signal as required, played it back, and stored it as an audio file.

2) Task 3: Transmitting the First Bit

- Modulation method: Our modulation method is PSK. We define a signal representing 0, and its opposite signal representing 1.
- Frame structure: Our frame structure includes preamble and data stream.
- Signal transmission: We detect the preamble of the signal by the magnitude of its correlation. When searching for the first preamble, a threshold is used, which means that if the size we set is reached, the preamble is considered to have been found. Afterwards, we search for the preamble based on the maximum correlation, and once we find the preamble, we decode the data to obtain the information.

C. Challenges

- Equipment Limitations: The performance differences between devices and the environment make the results complex.

- Modulation and Synchronization: The solution we initially implemented has weak noise resistance and is prone to misalignment errors during synchronization.

D. Solutions

- We have optimized the overall workflow. By adjusting the preview rating, it can be clearly distinguished from the data stream.
- Modulated the step size for searching for preambles to ensure that no other preambles are read during synchronization.
- Determined the correlation threshold required to find the first preamble by plotting the correlation between the entire signal and the preamble.

E. Suggestions

- Enhanced Resources: Looking back at Project 1, although it's not difficult, it took quite some time to get started. Including the implementation of the solution, the modulation parameters are also very critical. Moreover, Project 1 serves as the foundation for other projects. I suggest summarizing the key and difficult points of Project 1 in the future to help students understand the entire process step by step. Only by organizing Project 1 can it be flexibly applied in future projects. Otherwise, it will greatly dampen the students' enthusiasm for learning.

II. PROJECT 2: MANAGE MULTIPLE ACCESS

A. Stories

While Project 1 involves working on wireless transmission, Project 2 is our first time using sound card and cables to carry information between two nodes, i.e., two computers. In theory, this is more accurate and faster for transmitting data than not using wire connections. In order to connect the two nodes in hardware, we connected the computer, cables, and sound card according to the image in the PDF required by the project, which took us a long time.

B. Designs

Task 1 requires us to transfer the file INPUT.bin from NODE 1 to NODE 2 via Ethernet and decode it correctly to obtain the file OUTPUT.bin. We need to read the file on NODE 1, convert hexadecimal numbers to binary, and then convert these binary numbers into signals to be sent sequentially. NODE 2 listens to signals through cables, decodes them into binary digits, converts them into hexadecimal digits, and writes them to a file.

C. Challenges

- A hexadecimal number can be converted into 4 binary numbers. Also, in the BIN file, each time we read 2 hexadecimal numbers. That means, if one of the eight consecutive binary digits is inconsistent with the original during decoding, then the decoding of the entire digit in the BIN file will be incorrect. This will significantly reduce the accuracy of the final OUTPUT.bin file, and we have even encountered less than 1% accuracy.
- In addition, the question also has requirements for transmission time. The total recording time should be slightly longer than the playing time. Due to the varying performance of different computers, the start time of recording also needs to be manually adjusted.

D. Solutions

- As we said before, it took us too much time to implement Task 1. We tried a lot to let the parameters of the signals and the preamble to be suitable, so that the signals can be decoded more accurately. We chose preambles, which have a frequency range from 2000 to 8000 Hz and a frequency interval of 30 Hz. We chose a sine wave with a frequency of 12000 Hz as the signal and represented the binary digits 0 and 1 using PSK.
- The requirement for transmission time is easily resolved. After determining whose computer is NODE 1 and whose computer is NODE 2, we conducted multiple attempts to adjust and confirmed the time for each person to click and run the program.

E. Suggestions

- Project Complexity: Although we did not complete all the tasks, we also viewed the requirements of other tasks carefully and estimated the time and difficulty to complete them. We believe that project 2 is the most challenging among all the projects. We propose to give students more time to complete Project 2. Alternatively, some tasks from Project 2 could be moved to Project 1 or Project 3.

III. PROJECT 3: TO THE INTERNET

A. Stories

The part of "Stories" is narrated in the first person by Yunhao Shi.

Due to the previous two projects, there was no need for a computer to both play and record, so I haven't found that my computer can't record using the cable and the sound card. This seems to be a problem with my computer's USB driver, but until now, our team and TAs haven't accurately located the problem.

We tried replacing the sound card, cables, and even connecting cables to another USB port on my computer, but all of these failed. We tried many methods, but none of them worked. My teammate put the code on the computer of his roommate who didn't participate in the CS120 course and found that

his roommate's computer could record normally. This further proves that it is a problem with my computer equipment.

It is also impossible for us to buy a new computer just for this course or borrow someone's computer for a long time. So we tried to divide each task into at least 2 parts, ensuring that my computer was always the playing node and my teammate's computer was always responsible for recording. To meet the time requirements of each task, we also recorded and sent the completion time when converting devices, and finally added up all the time to obtain the completion time of the entire task. Unfortunately, the TAs didn't approve of our solution. Although we have completed all tasks from 1 to 3, we were unable to test even one task during the check.

Fortunately, when my computer is recording, it will output the received sound as a very weak sound. Although neither we nor the TAs can understand this at all, it is just a coincidence and strangeness. Taking advantage of this property, we re-implemented Task 1 and checked with one TA on Tuesday night, and we managed to get some scores.

B. Designs

In the final version where we re-implemented Task 1, we chose Yunhao Shi's computer as NODE 1 and Tinghua Wen's computer as NODE 2. Based on Project 2, we added new functions as follows:

- The computer searches for the preamble through correlation during receiving and immediately records the corresponding signal once it is found.
- After recording a signal, it automatically plays the signal that needs to be returned.
- After playing the signal, it enters the receiving state again.

Each frame includes a preamble, IP, and data. This facilitates determining whether the corresponding frame has been found during decoding and ensures correct decoding.

We also utilized the functionality of socket packets to achieve network transmission of IP addresses.

C. Challenges

The part of "Challenges" is narrated in the first person by Yunhao Shi.

- First Challenge: My computer couldn't receive signals correctly. We have verified that the entire receiving and decoding process is fine and can run correctly on other devices, but only my computer can't receive signals correctly.
- Second Challenge: In previous projects, we used one device to play signals and another device to receive signals. Then we will search for the preamble in the recorded signal. Now we need a device that is constantly listening and automatically switches to playing state when it detects a preamble and the signals of one frame. Similarly, after playing the signal on another device, it should automatically switch to receiving state.
- Third Challenge: The question is how the entire ping process is implemented when three nodes are connected.

This mainly includes what information each node needs to propagate and receive.

D. Solutions

- For the first challenge, we have no good solution. We used a very tricky approach to solve it, but the effect was limited.
- The second challenge is very complicated, but it is relatively easy to implement by using loops.
- In our previous attempts, the third challenge was not difficult either. Proper use of socket packages can help us quickly and accurately implement requirements.

E. Suggestions

Because this project is closely related to the equipment. Once there is a problem with the equipment, the difficulty of completing the project will increase. Borrowing or buying equipment is also a very troublesome thing. We did not find the process to apply to use the computer in the data center because we had already passed the deadline when we learned that our practical plan could not be used.

It is suggested that if there are students with equipment problems in the future, this course can guide them to apply for data center testing. Or the students can communicate with the TAs early to find other solutions.

IV. PROJECT 4: ABOVE IP

A. Stories

At first, we didn't quite understand the meaning of the question. We don't know how to resolve domain names using a given IP address. For this reason, we even searched online for the meaning of the IP 1.1.1.1 and learned that it was the result of Cloudflare's collaboration with APNIC. Sending the domain name to this IP can obtain the corresponding IP address. Meanwhile, 1.1.1.1 is also one of the fastest parsing servers in the world.

Due to time and device limitations, we have abandoned Task 2.

B. Designs

On the basis of Project 3, we have designed a method for transmitting string information from NODE 1. NODE 1 converts the input domain name string into a signal and transmits it to NODE 2 via cable. After decoding the signal into a domain name, NODE 2 sends the domain name to 1.1.1.1 using socket packets and receives the IP address sent from 1.1.1.1. Subsequently, NODE 2 sends the obtained IP to NODE 2.

C. Challenges

- The biggest challenge is still that the recording process on Yunhao Shi's computer can't work well. In this task, we found that the correlation of the signal is only about one tenth of the correlation in Project 3. Such a small correlation makes it almost impossible for us to distinguish between preamble and noise, let alone extract the digits of the IP address and decode them correctly.

- There is still a challenge. Unlike in previous projects where numbers were transmitted between NODE 1 and NODE 2, in Project 4 we need to transmit strings. If the decoding is incorrect, it will turn into abstract byte type characters instead of numbers, which poses significant difficulties for our debugging work.

D. Solutions

- We tried a lot, but we failed. So we can only try to program without test. We also found a student majoring in chemistry and borrowed his computer to test the code. But due to time constraints and personal reasons, we can only borrow his computer and try debugging on the day of the check.
- We have no choice but to try multiple times to fine tune the parameters of preamble and correlation. We have gone through nearly 100 attempts.

E. Suggestions

Same as Project 3. We have no other suggestions.